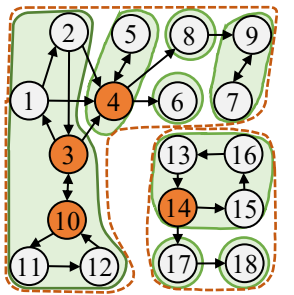
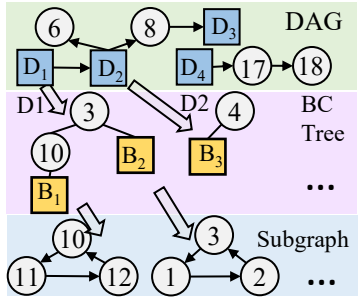


(a) An example graph



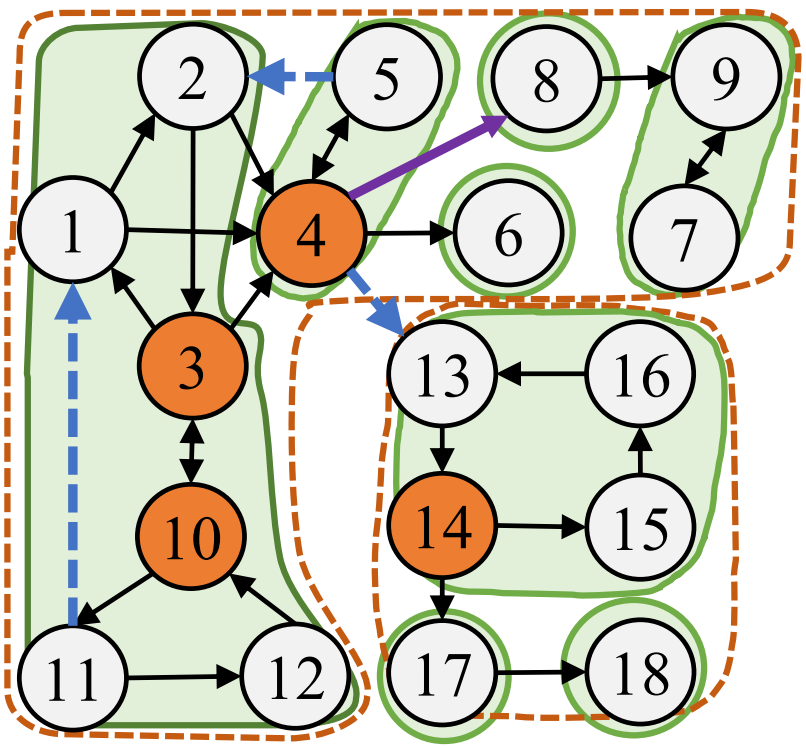
(b) Find WCC, SCC, BiCC

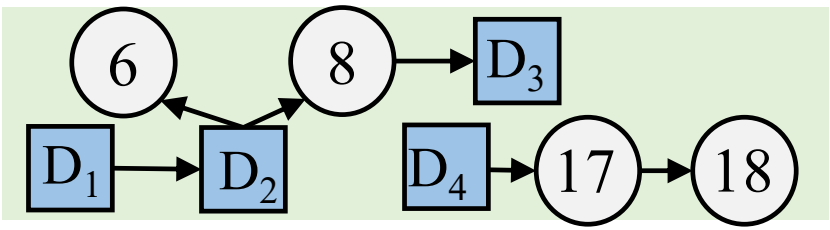


(c) GIR Construction

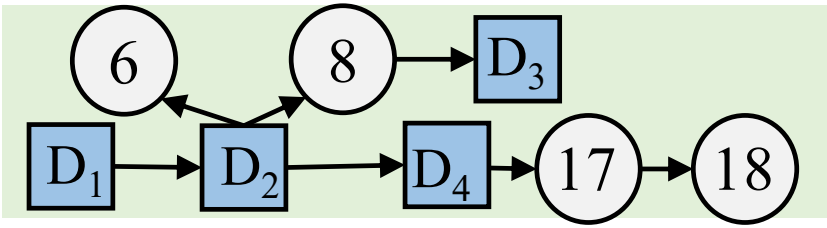
Graph Computation	
Traversal	*BFS* *DFS*
Path finding	*SSSP* *KSP* *APSP*
Centrality	*BC* *CC*

(d) GIR Application

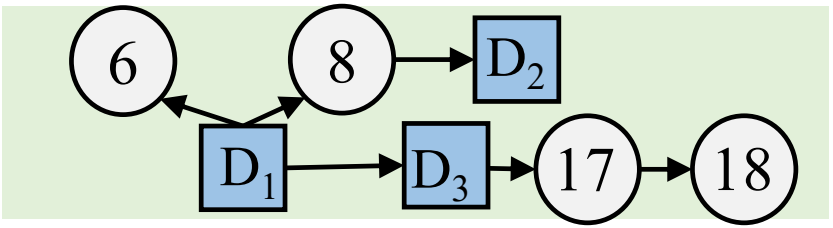




Case 1: different WCC: (4, 13)



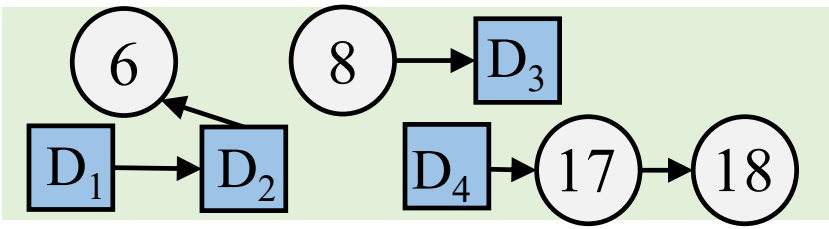
Case 2: different SCC: (5, 2)

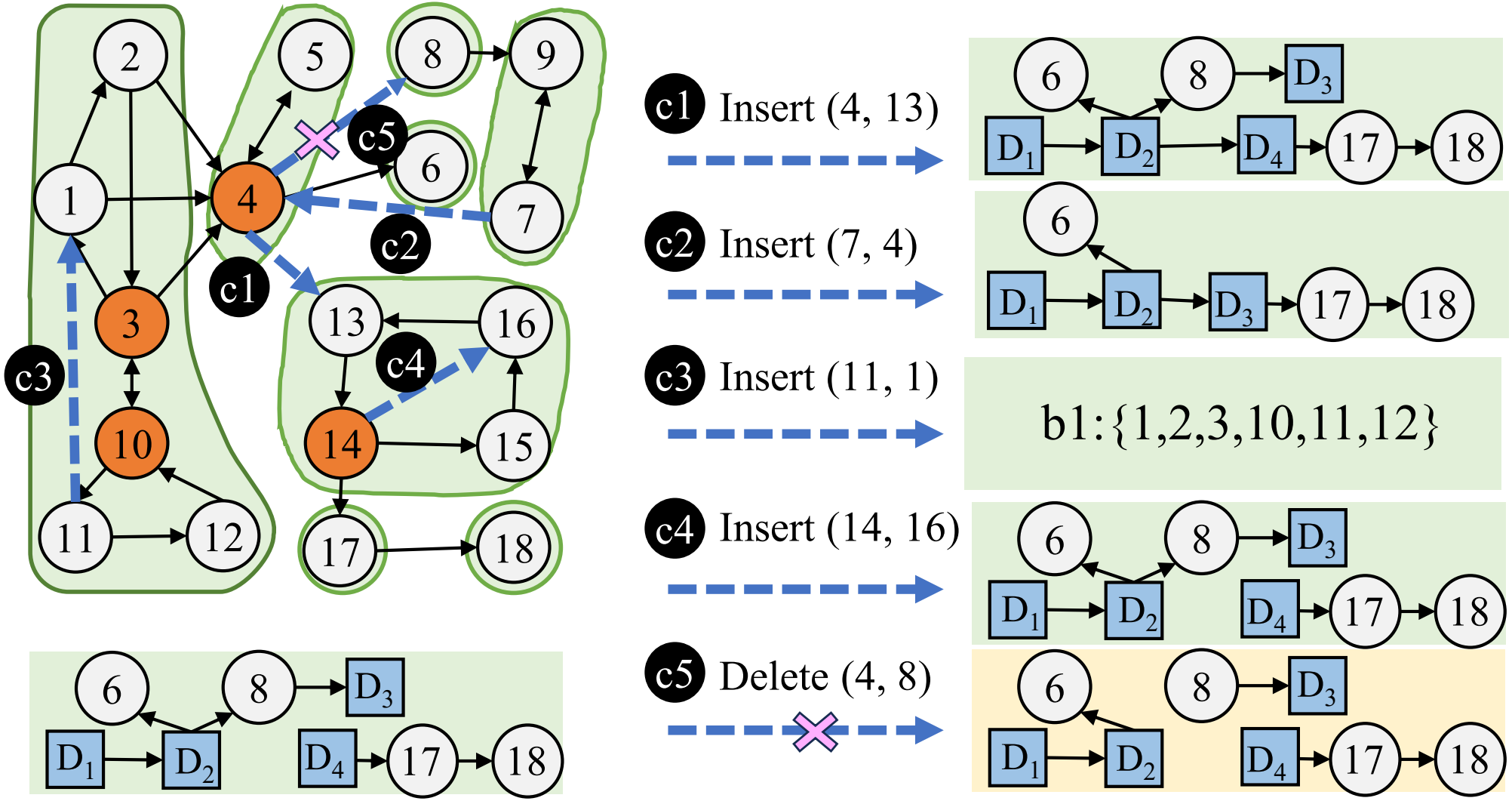


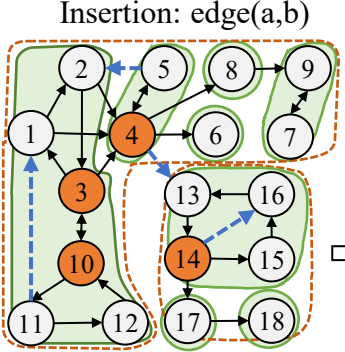
Case 3: different BiCC: (11, 1)

b1:{1,2,3,10,11,12}

Case 4: bridge(4, 8)




c1 Insert (4, 13)
c2 Insert (7, 4)
c3 Insert (11, 1)
c4 Insert (14, 16)
c5 Delete (4, 8)
b1:{1,2,3,10,11,12}
(a) original graph and GIR
(b) Updated graph and scope

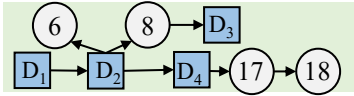


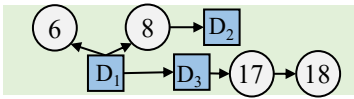
Case 1: different WCC: (4, 13)

Case 2: different SCC: (5, 2)

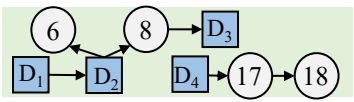
Case 3: different BiCC: (11, 1)

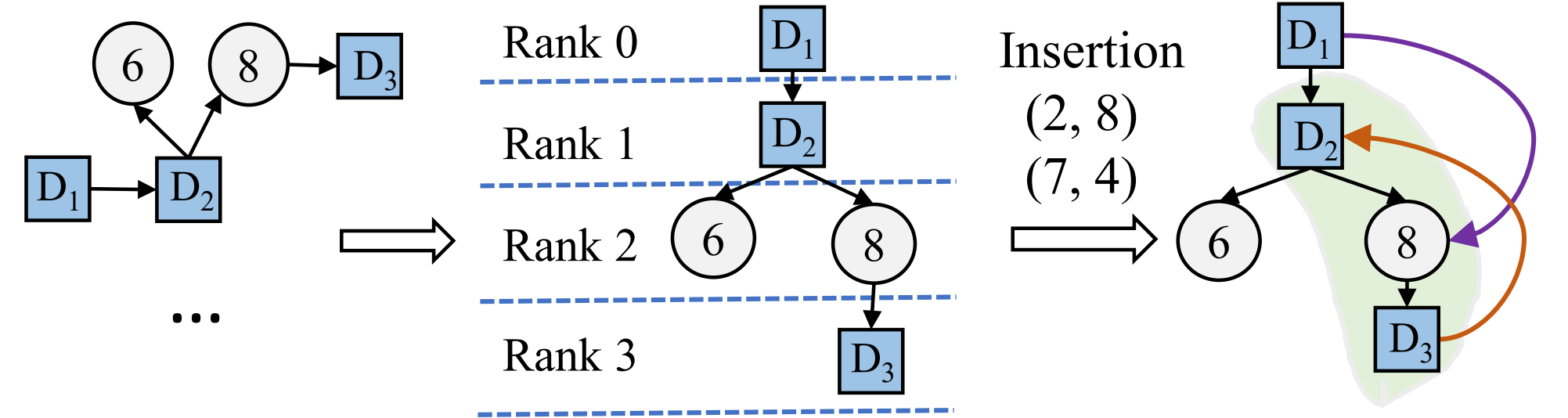
Case 4: same BiCC: (14,16)



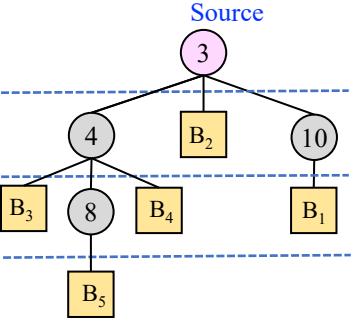


b1:{1,2,3,10,11,12}

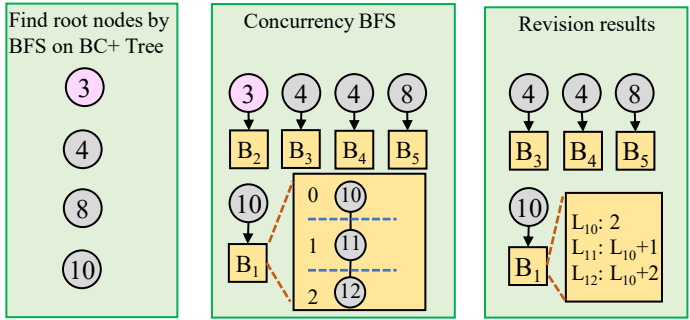




(a) Example graph (b) Node rank (c) Scope-prune-update



(a) Example graph



(b) Detailed workflow